

Listening to Southern Maine

A Data-Driven Look at Community Challenges and Civic Engagement

United Way of Southern Maine × Northeastern University

- 1,700+ community voices across Cumberland & York Counties
- Mixed-methods: survey analysis, NLP classification, statistical modeling
- Who is struggling, who feels unheard, and how to engage them



About the Data: Southern Maine Community Survey

Respondents from Cumberland and York Counties | Community Survey + ALICE Voices

1,702

TOTAL RESPONDENTS

Cumberland & York Counties

65% / 30%

CUMBERLAND / YORK

4% listed both

30%

BELOW ALICE THRESHOLD

Struggling to afford basics

74%

FEEL THEIR VOICE IS UNHEARD

Answered 'Yes' or 'Maybe'

19%

WANT TO GET INVOLVED

Ready to be part of the solution

Understanding the Data: Three Survey Types

Three surveys merged into one dataset — subgroup analyses restricted to relevant survey types

Abbreviated Survey

n = 985 respondents

- ALICE Status
- Location
- Biggest Challenges
- Feel Heard
- Age / Race
- UWSM Relationship
- Want Involvement
- Community (Unheard)
- Hardest Expenses
- Employment
- Roadblocks
- Helpful Supports

Full Survey

n = 639 respondents

- ALICE Status
- Location
- Biggest Challenges
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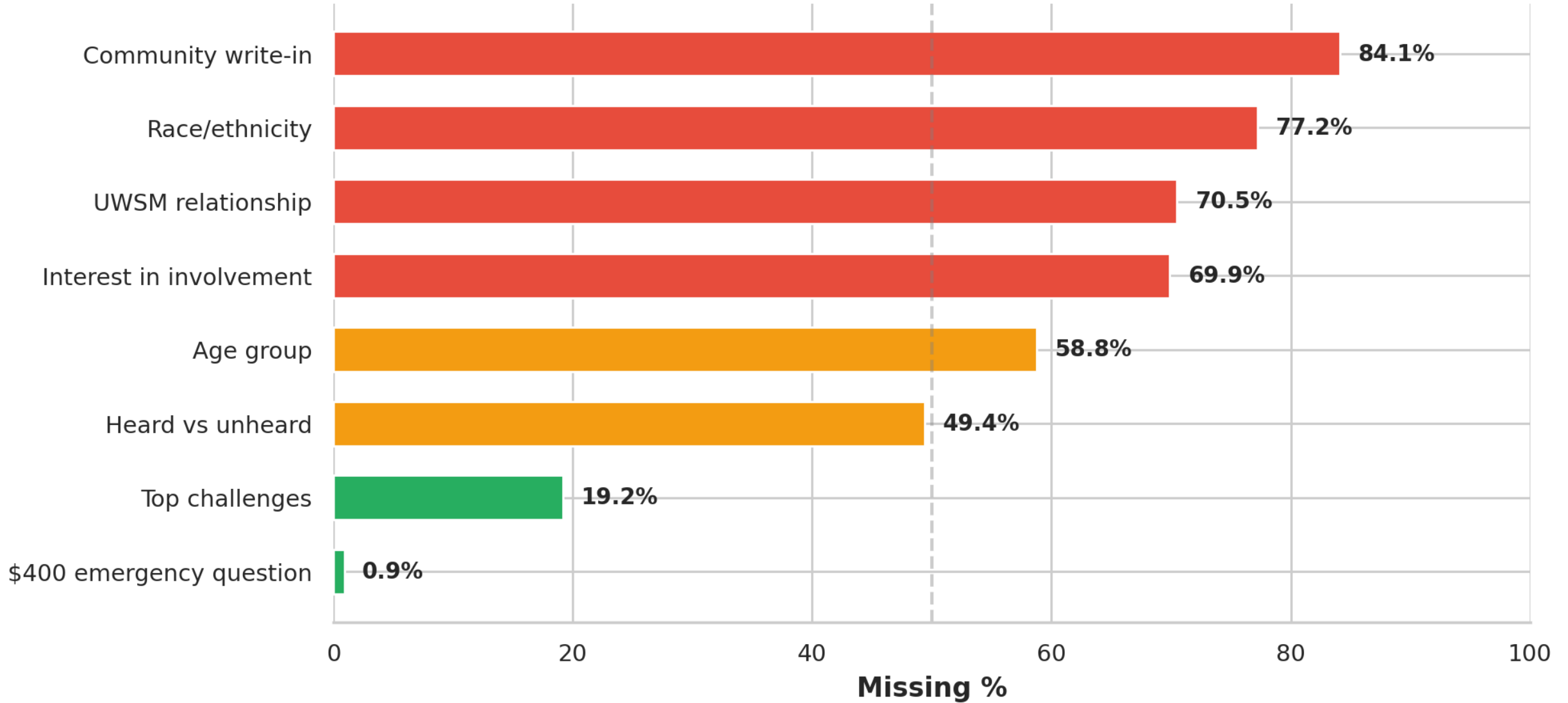
ALICE Voices

n = 78 respondents

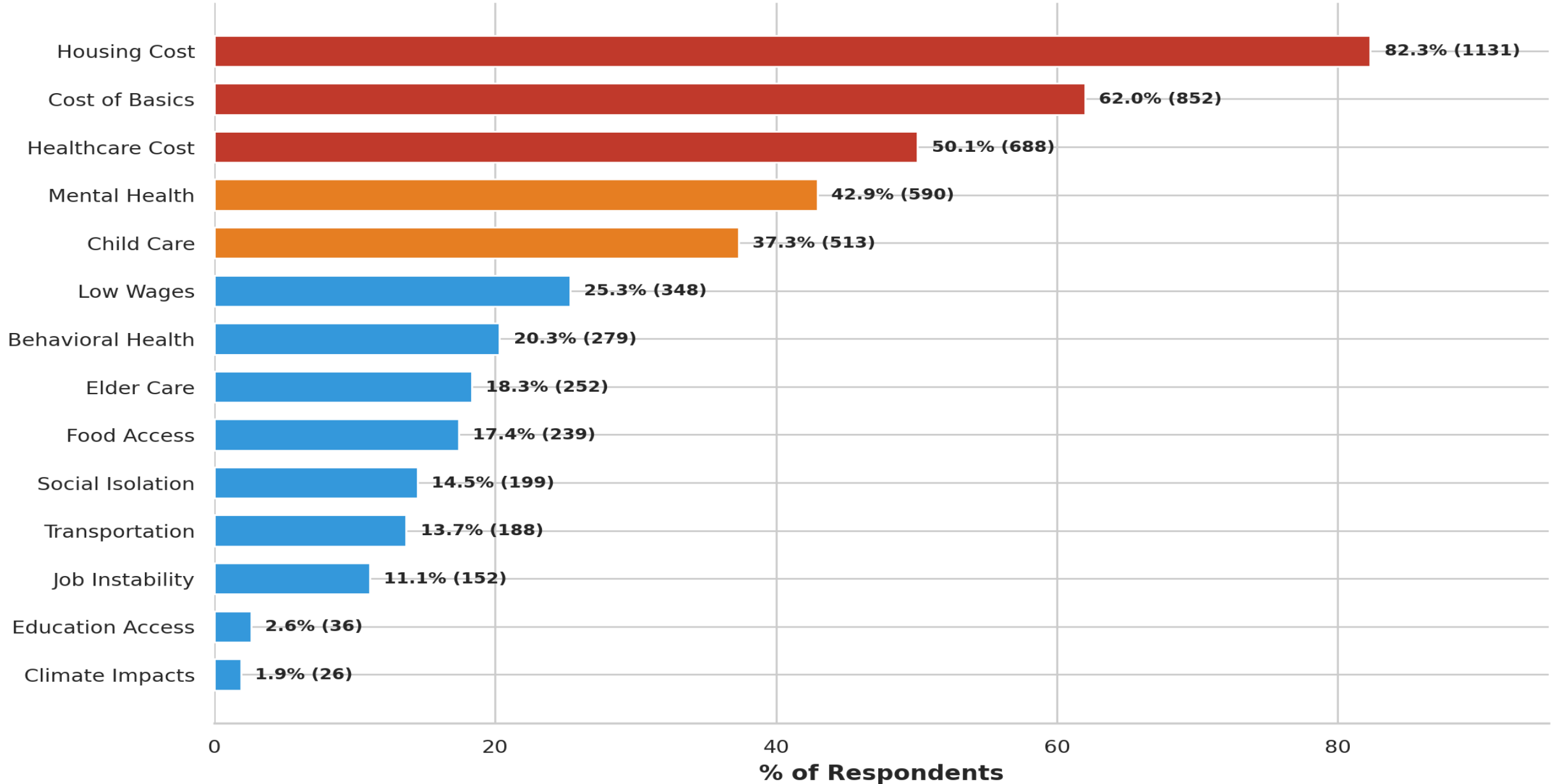
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Data Completeness: What We Can Trust

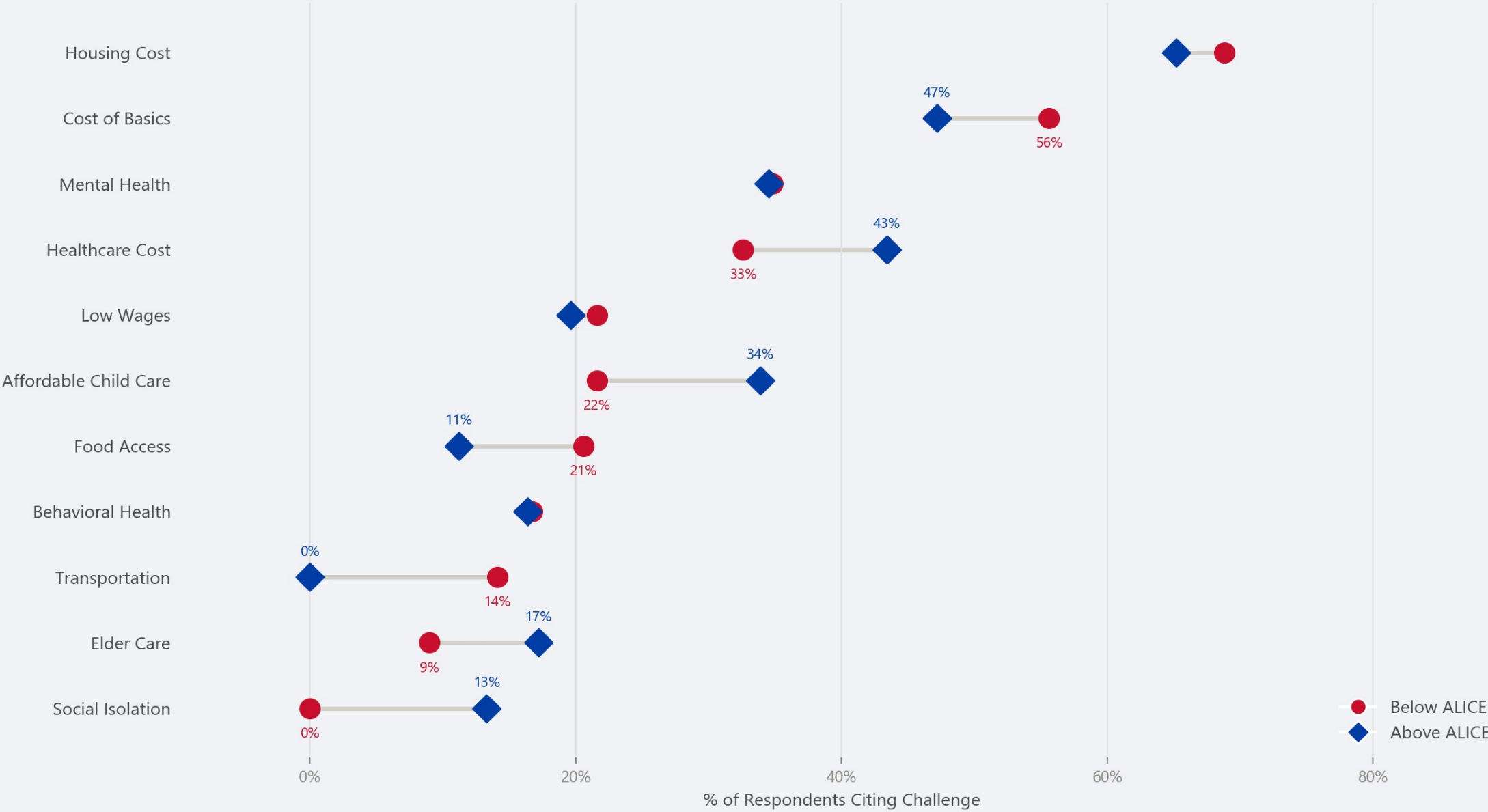
Data Completeness Across Key Fields



Top Community Challenges Overall (n=1374)



Community Challenges: Below ALICE vs. Above ALICE Households



Community Challenges Mapped to Maslow's Hierarchy

Physiological Needs

Housing Cost (82%) | Cost of Basics (62%) | Food Access (17%)

Safety & Security

Healthcare Cost (50%) | Low Wages (25%) | Job Instability (11%) | Transportation (14%)

Belonging

Social Isolation (14%) | Behavioral Health (20%)

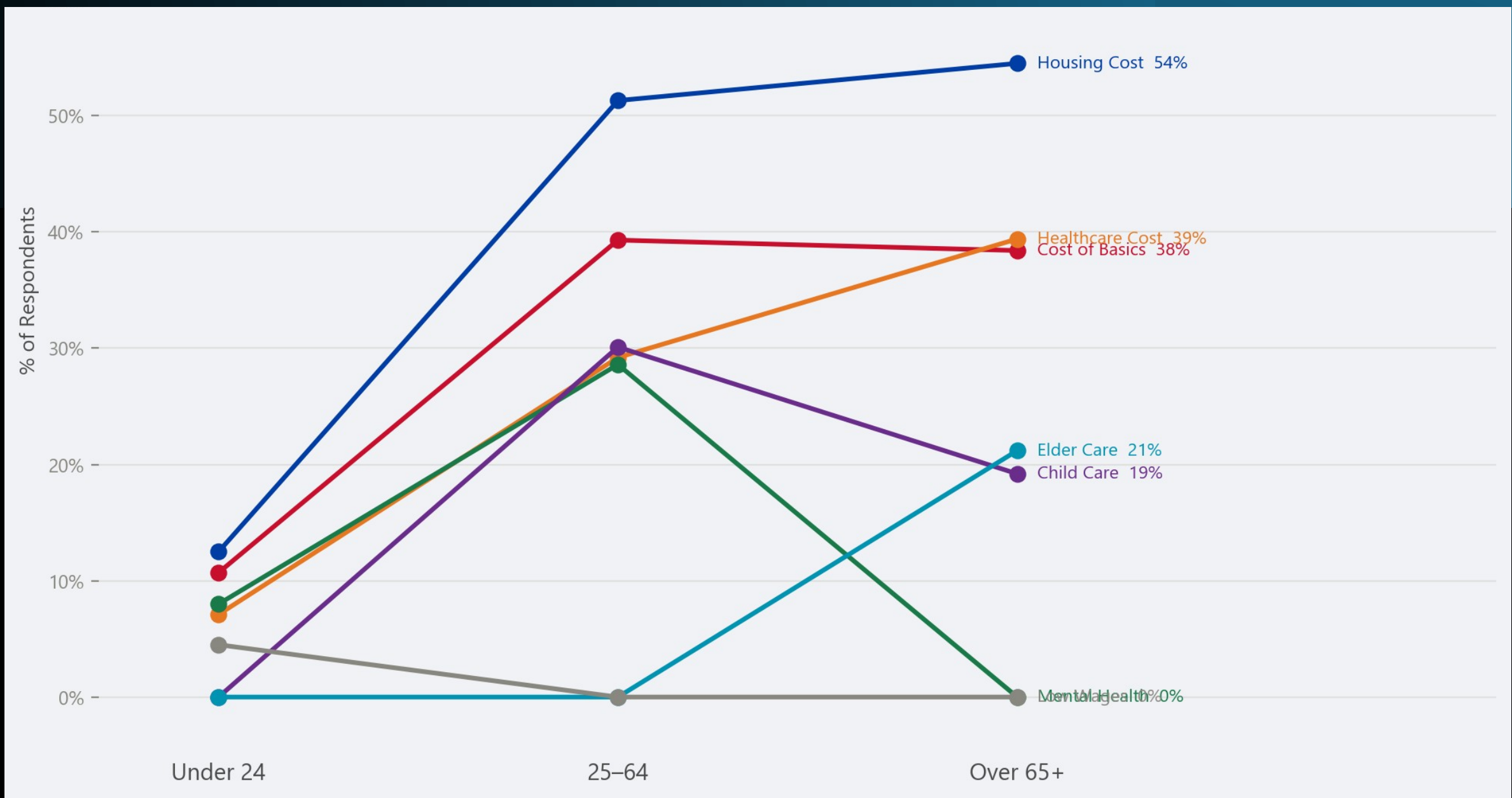
Esteem & Growth

Education Access (3%) | Child Care (37%) | Elder Care (18%)

Self-Actualization

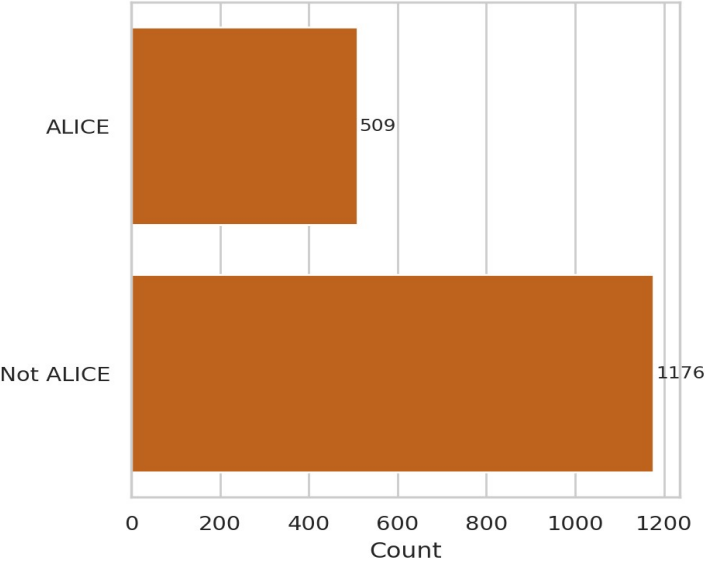
Climate Impacts (2%)

Base needs (housing, food, basics) dominate — higher-level challenges are less reported

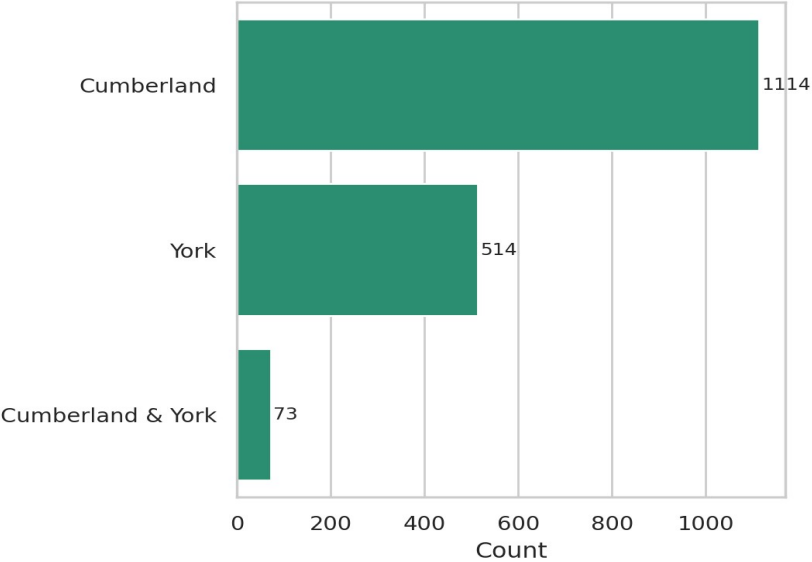


Respondent Profile — Core Cumberland/York Sample

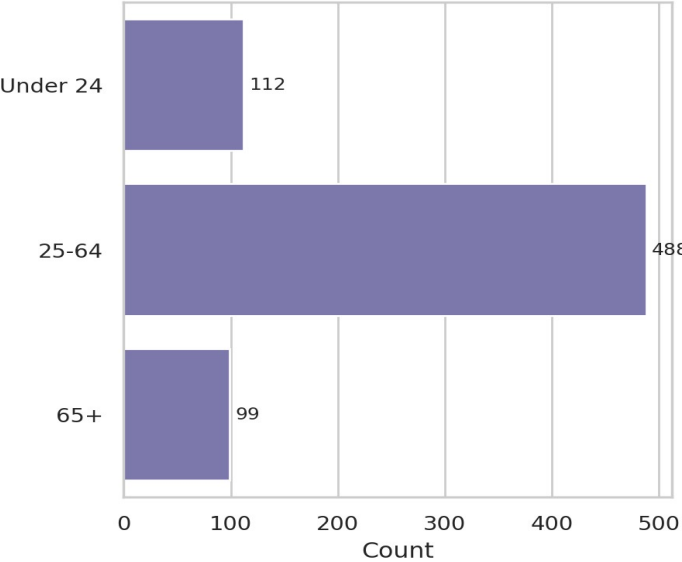
ALICE Proxy (n=1685)



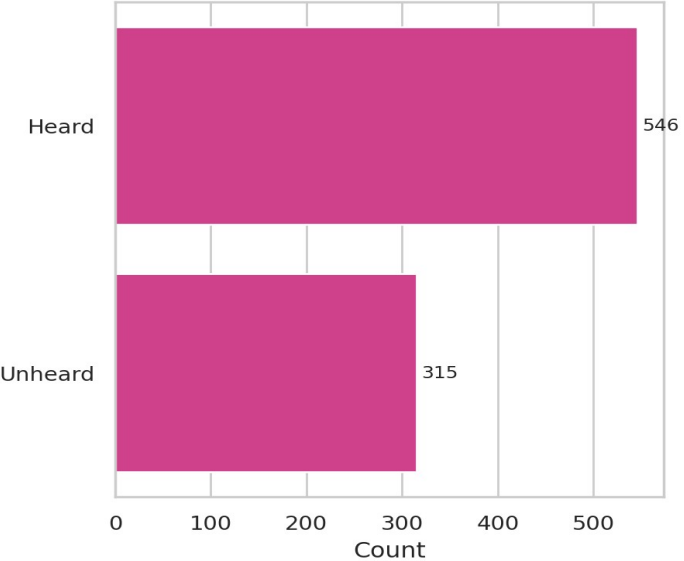
County (n=1701)



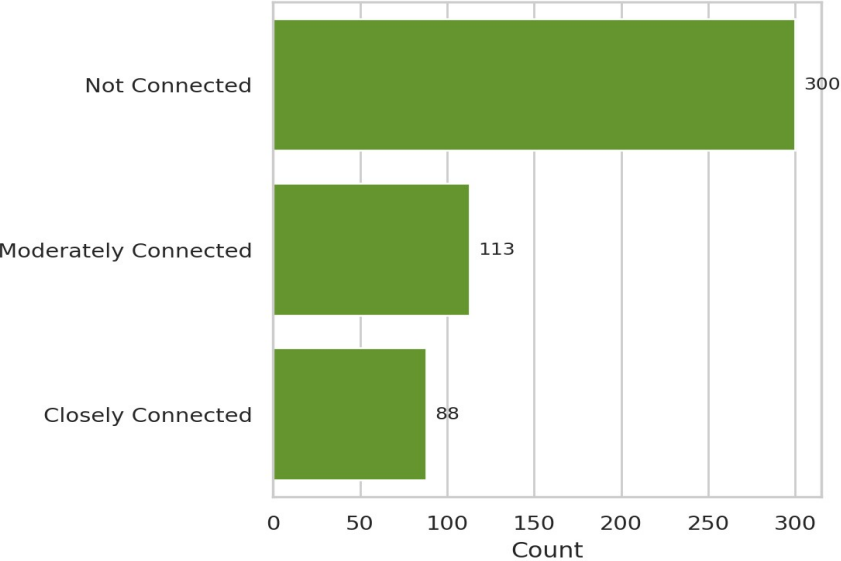
Age Tier (n=699)



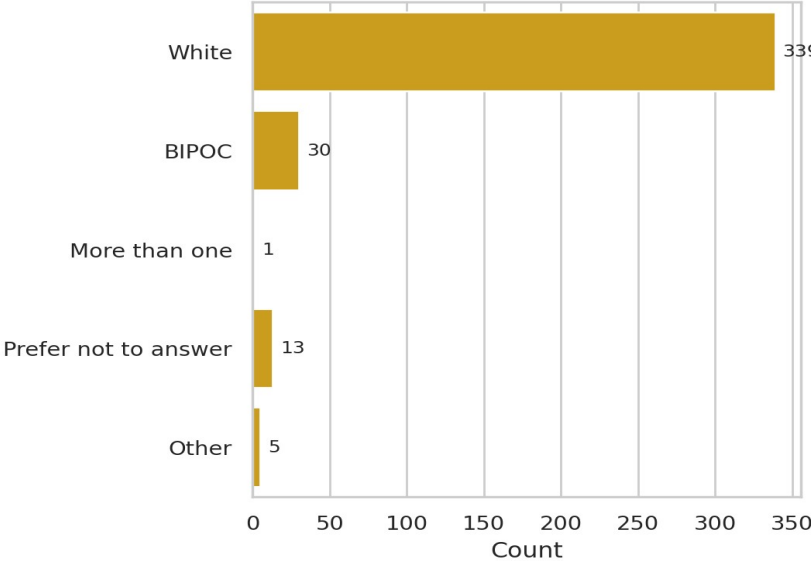
Heard vs Unheard (n=861)



UWSM Relationship (n=501)



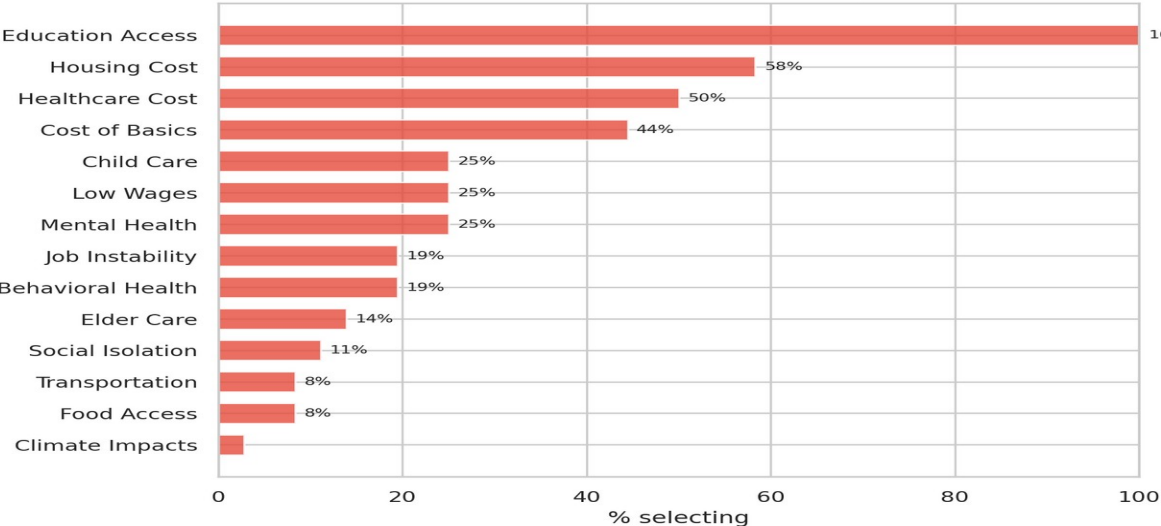
Race Group (n=388)



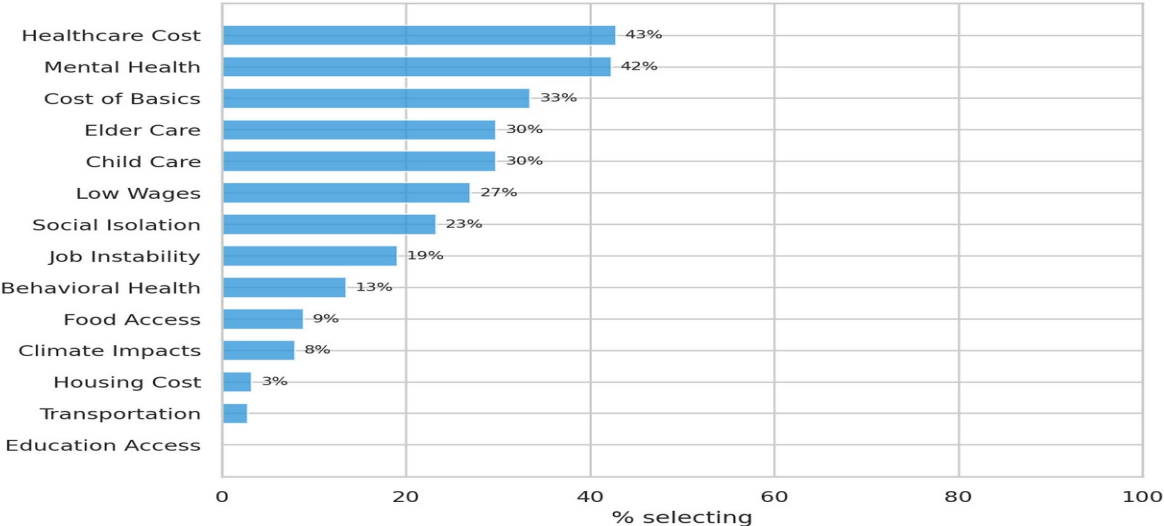
K-Means Clustering: Respondent Segments

K-Means Clustering: 4 Respondent Segments

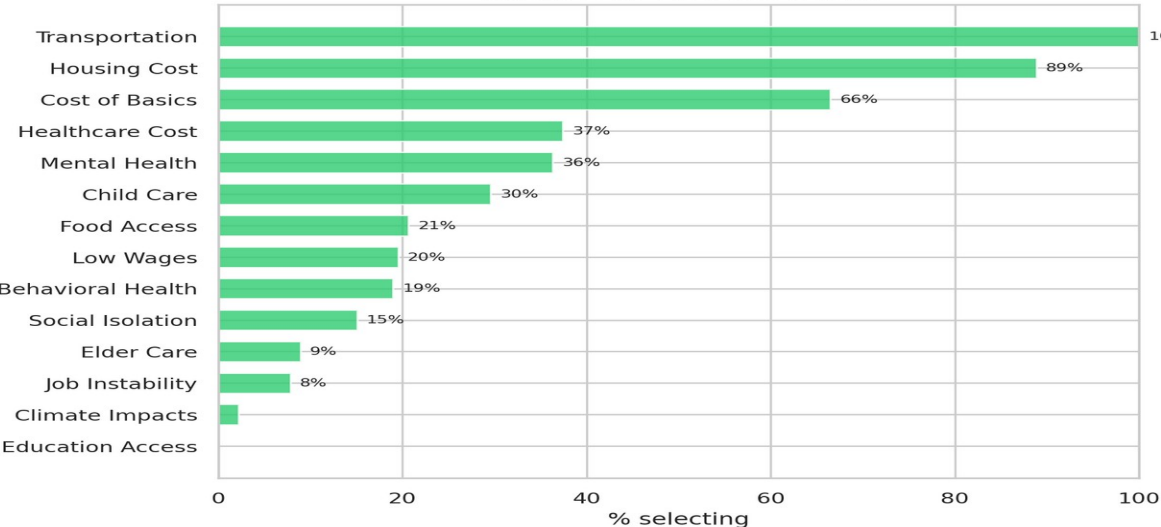
Cluster 1 (n=36, 49% ALICE)



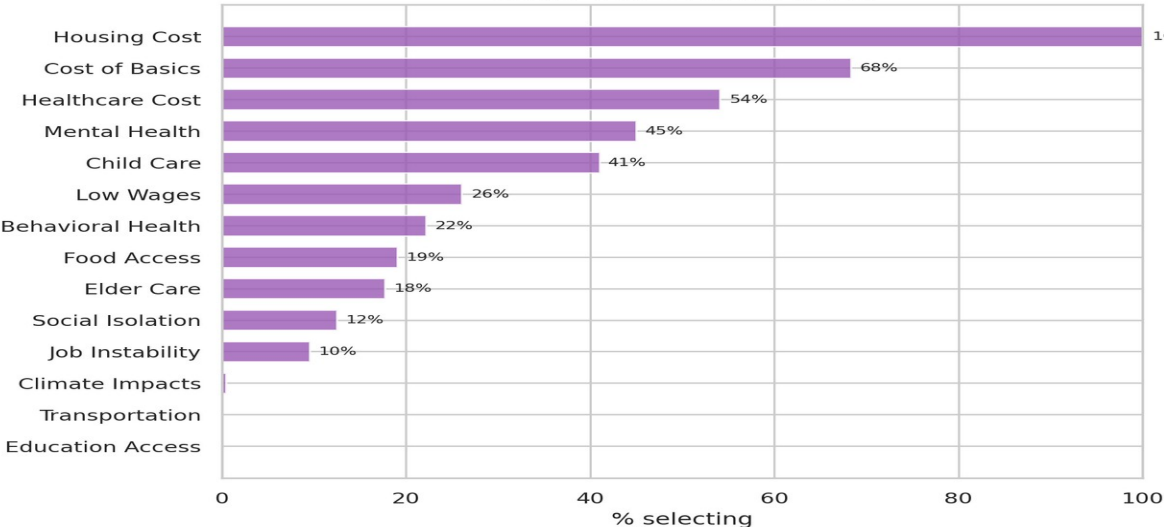
Cluster 2 (n=215, 25% ALICE)



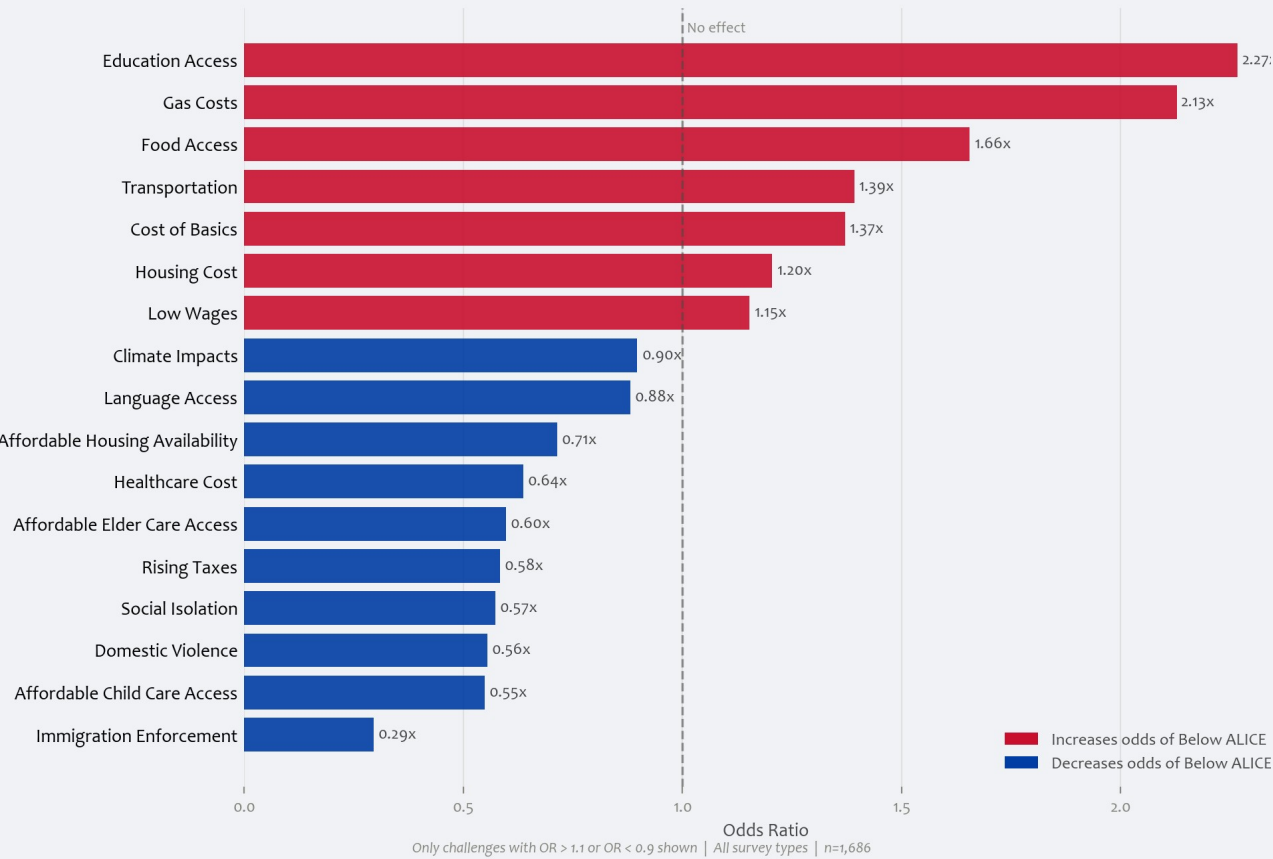
Cluster 3 (n=179, 40% ALICE)



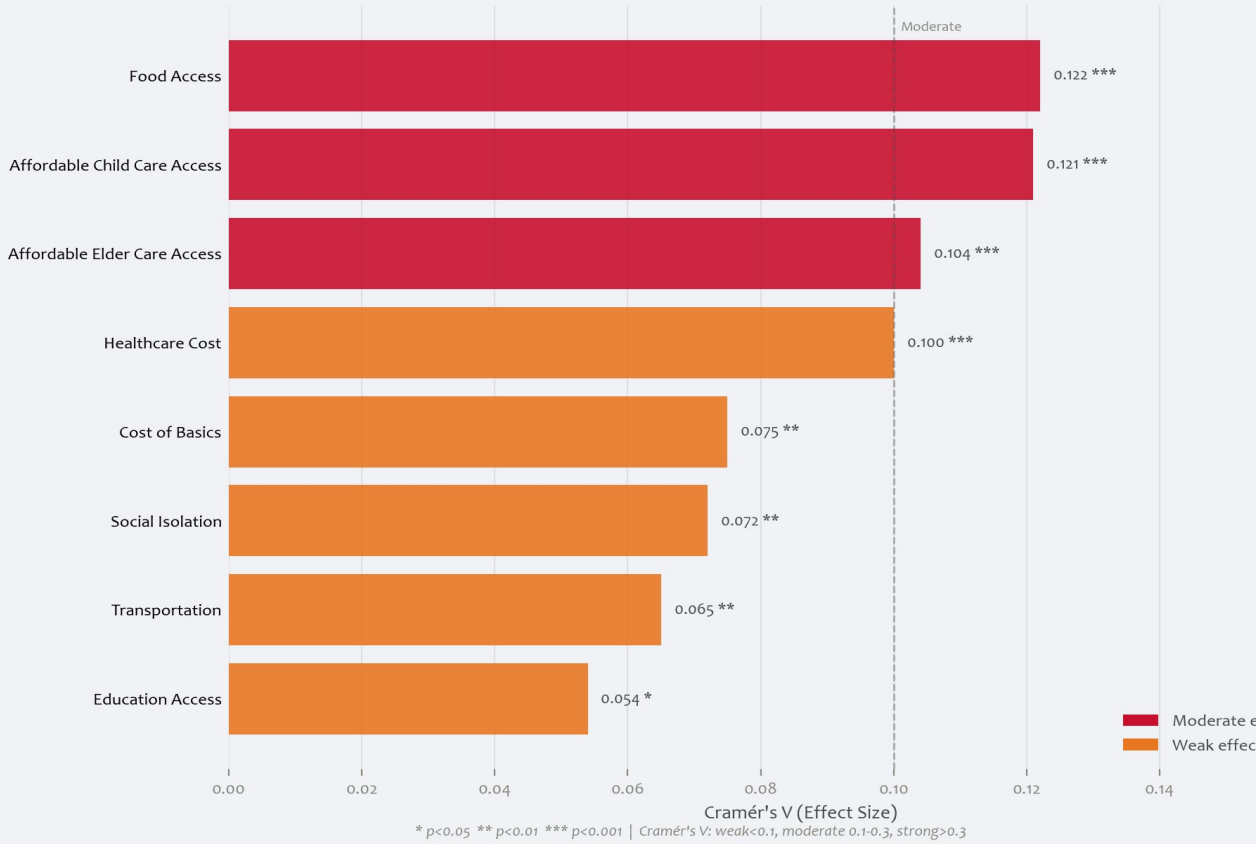
Cluster 4 (n=944, 29% ALICE)



Which Challenges Predict ALICE Status?
(Logistic Regression Odds Ratios)



Statistical Strength of Challenge Differences
by ALICE Status (p < 0.05 only)



Problems We Encountered

► High Missing Data

Demographics (age, race, UWSM) are 59-84% missing — limits subgroup reliability.

► Survey Type Mismatch

3Q respondents only answered 3 items, inflating missingness in other fields.

► Racial Imbalance

Only 30 BIPOC vs. 339 White — too small for robust race-based analysis.

► PCA Spread

First 2 components explain only ~27% of variance — challenges don't reduce cleanly.

Recommendations for UWSM

► Incentivize Completion

Offer incentives to boost demographic response rates from ~30% to 70%+.

► Finer Geography

Collect neighborhood/census-tract data beyond county for hyperlocal targeting.

► Housing is #1

82% cite housing — prioritize rent relief, affordable housing, housing navigation.

► Bundle ALICE Services

ALICE households need food, transport, and childcare together — wrap-around model.

► Mental Health Integration

43% cite mental health — embed behavioral support into financial programs.